HAMILTONIAN ELEMENTS IN ALGEBRAIC K-THEORY

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Abstract. A Hamiltonian bundle $M \hookrightarrow P \to X$ (with monotone compact fibers) induces a kind of "bundle of A_∞ categories" over X, with fiber given by the Fukaya category of M. Algebraic K-theory ideas, the above picture for $X = S^m$, and basic representation theory yield the following: if R is a regular ring and G is a compact Lie group then there is a natural group homomorphism $\pi_m(BG) \to K_m(R) \oplus K_{m-1}(R)$. Our construction naturally first assigns elements in a certain categorified algebraic K-theory $K_m^{Cat}(R)$. And the construction yields that $K_2^{Cat}(\mathbb{Z})$ is infinitely generated (with details postponed). This is in contrast to finite generation of standard algebraic K-theory of $\mathbb{Z}$ already proved by Quillen. This categorified algebraic K-theory is analogous to Toën's secondary K-theory of R. Taking the Langlands dual of G, we explore the relationship between the images of the corresponding homomorphisms above.

1. Introduction

A guiding idea is that a Hamiltonian bundle $M \hookrightarrow P \to Y$ gives rise to a kind of "bundle of A_∞ categories", with fiber the Fukaya category of M. Precisely what this means is an interesting mathematical problem in its own right, and one answer is developed in [17]. Note that we are not talking about local systems, as the "bundles" that one gets are generally non-trivial over spheres as shown in [16].

The second idea, due to Toën [20], is that there is a refinement of K-theory of a commutative ring R, by replacing R-modules with dg or A_∞ categories over R, and using Waldhausen K-theory of a commutative ring. We show that these two ideas naturally fit together. We use the above assignment, with $Y = S^m$, to associate to P elements in categorified and standard algebraic K-theory. This culminates in a connection between symplectic geometry and algebraic K-theory, and in particular we get a certain geometric insight into what we call categorified algebraic K-theory of $\mathbb{Z}$. For example, we obtain an infinite generation result. At the end we explore connections with Langlands duality, with the goal of getting a new K-theoretic expression of the electric-magnetic duality.

Although in the main examples here the principal actors are compact groups, in particular $PU(n)$, the framework is most naturally understood from the perspective of the group of Hamiltonian symplectomorphisms of some symplectic manifolds; so we quickly introduce this. Let (M, ω) be a symplectic manifold, throughout assumed to be compact, with ω a closed nondegenerate 2-form on M. The group of diffeomorphisms ϕ of M, satisfying $\phi^*\omega = \omega$, is called the symplectomorphism group of (M, ω). When M is simply connected, this is also the group of Hamiltonian symplectomorphisms, which we denote by $\mathcal{H} = \mathrm{Ham}(M, \omega)$. In general, a symplectomorphism is Hamiltonian if it is the time-one map of an isotopy generated by a family of vector fields $\{X_t\}$, $t \in [0, 1]$, satisfying

$$\omega(X_t, \cdot) = dH_t(\cdot),$$

for some family of smooth functions $\{H_t\}$ on M.

The group $\mathcal{H}$ is one of the more enigmatic objects of mathematics. It is always (unless $M = pt$) an infinite-dimensional Fréchet Lie group with its C^∞ topology. But it has a natural, bi-invariant

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Finsler metric called the Hofer metric. So in some ways it behaves like a compact group. Its topological and algebraic structure is tied to the main problems of Hamiltonian dynamics, see for instance Polterovich [12].

Much of the algebraic/topological structure of $\mathcal{H}$ is reflected in the properties of Hamiltonian fibrations, which are the main geometric ingredients in what follows. A *Hamiltonian fibration* $M \hookrightarrow P \to Y$ is a smooth fiber bundle over Y, with fiber a symplectic manifold (M, ω), whose structure group is contained in $\mathrm{Ham}(M, \omega)$. For example, take $(M, \omega) = (\mathbb{CP}^n, \omega_{FS})$ where ω_{FS} is the Fubini-Study symplectic form, and take the structure group $G = PU(n+1) \subset \mathrm{Ham}(\mathbb{CP}^n, \omega_{FS})$.

The other ingredient is algebraic K-theory, which is a construction associating an infinite loop space [1] $K(R)$ to any ring R , and so abelian groups:

$$K_m(R) = \pi_m(K(R)), \quad m \in \mathbb{N},$$

called K-theory groups of R. This is a powerful tool in algebraic topology and ring theory, with applications in number theory and geometric topology.

One motivating example for algebraic K-theory is the following. Take R to be the non-commutative ring $\mathcal{K}$ of compact operators on a separable complex Hilbert space, the algebraic K-theory space $K(\mathcal{K})$ [2] is identified with the topological K-theory space $KU = BU \times \mathbb{Z}$, see Karoubi [6], Suslin-Wodzicki [19]. In particular, a complex vector bundle E over a space X determines a homotopy class $[f_E : X \to K(\mathcal{K})]$, which we might call a *geometric element* of the corresponding abelian group. We shall see that in a certain sense this extends to a general ring R, provided it is commutative, and replacing a complex vector bundle by a suitable Hamiltonian fibration.

For each commutative ring R, we construct "categorified algebraic K-theory groups" $K_m^{Cat}(R)$. Actually it will be convenient to define more general groups $K^{Cat}(X, R)$ for each simplicial set X, generated by aforementioned bundles of A_∞ categories over X with fibers A_∞ categories with finite type, see Definition 3.1. This is analogous to the group of classes $[Y \to K(\mathcal{K})]$ for a space Y, i.e. the topological K-theory of Y.

In principle, this categorified algebraic K-theory of R is a slight variation of the secondary K-theory of R introduced by Toën, with the main difference that $\mathbb{Z}$-graded chain complexes are replaced by $\mathbb{Z}_2$-graded chain complexes. By way of the Hochschild chain functor, $K_m^{Cat}(R)$ should admit a natural map to the 2-periodic variant of algebraic K-theory of R, which we denote by $K_m^{\mathbb{Z}_2}(R)$, see Remark 3.7, although we don't yet need this.

Remark 1.1. It was pointed out to me by Bertrand Toën that for R a regular ring, for example $R = \mathbb{Z}$, $K_m^{\mathbb{Z}_2}(R) \simeq K_m(R) \oplus K_{m-1}(R)$. This can be proved with A^1 homotopy theory.

To obtain "geometric elements", we start with a Hamiltonian fibration $M \hookrightarrow P \to Y$, with fiber a monotone symplectic manifold. The following is a conjunction of Theorem 4.3 and Corollary 4.6.

Theorem 1.2. *If the Fukaya category of M is finite type, then for each R the data of the isomorphism class of P determines an element in $K^{Cat}(Y_\bullet, R)$. In particular, this gives mappings*

$$\phi_m : \pi_m(B\,\mathrm{Ham}(\mathbb{CP}^n, \omega_{FS})) \to K_m^{Cat}(R),$$

and so mappings

$$\phi_m : \pi_m(BPU(n) \to K_m^{Cat}(R).$$

Example 1. In the case of $n = 1$, in [16] using geometric arguments we outline the proof of non-triviality of g_4 for $R = \mathbb{Z}$. This contrasts with a theorem of Rognes [14] on the triviality of $K_4(\mathbb{Z})$.

[1] In this paper, the term infinite loop space is meant to be synonymous with connective Ω-spectrum, i.e. a sequence of spaces $\{E_n\}_{n\in\mathbb{N}}$ with $E_n \simeq \Omega E_{n+1}$. Then $K(R)$ is the notation for the 0-space E_0 of the corresponding spectrum.

[2] Rather one must take a certain non-connective variant of the algebraic K theory construction.

Claim 1.3. *$K_2^{Cat}(\mathbb{Z})$ is infinitely generated.*

The proof of this uses the mappings g_2, and certain symplectic geometry ingredients like the Seidel homomorphism, to construct group injections

$$\mathbb{Z}_p \simeq \pi_2(BPU(p)) \to K_2^{Cat}(\mathbb{Z})$$

for all prime p. The details will appear elsewhere. Since the groups $K_m(\mathbb{Z})$ (and hence $K_m^{\mathbb{Z}_2}(\mathbb{Z})$ assuming Remark 1.1) are always finitely generated by foundational results of Quillen [13], the latter conclusion is perhaps surprising.

Our construction also gives the following, which in principle lifts to categorified algebraic K-theory, see Section 4.1.

Theorem 1.4. *For any compact Lie group G, there is a homotopy natural map:*

$$h : \mathrm{BG} \to K^{\mathbb{Z}_2}(R), \tag{1.5}$$

and so there are natural homomorphisms

$$h_m : \pi_m(\mathrm{BG}) \to K_m^{\mathbb{Z}_2}(R).$$

Taking the Langlands dual of G, we explore correspondence of the respective mappings in Section 6.

Question 1. Can one obtain the classes

$$[\mathrm{BG} \to K^{\mathbb{Z}_2}(R)]$$

by techniques of homotopy theory and algebraic geometry? In other words, can we bypass the geometric analysis of the construction?

2. Preliminaries

We will mostly follow Keller [7], but our base commutative ring will be denoted R to avoid overloading notation. An A_∞ category C over R is a form of non-associative dg-category, with $hom_C(a, b)$ a cochain complex over R. We refer the reader to standard references for further details, see for instance [11] which will come in use later.

Denote by $A_\infty Cat_R^{strict}$ the category of strictly unital, $\mathbb{Z}$-graded A_∞ categories with morphisms strict unital A_∞ functors. We set $\mathrm{A_\infty Cat_R^{\mathbb{Z}_n,strict}}$ to be like $A_\infty Cat_R^{strict}$ but with chain complexes $\mathbb{Z}_n$-graded instead of $\mathbb{Z}$-graded. This means that all structure maps, functors, etc., are with respect to the $\mathbb{Z}_n$-grading. Various subscripts R may be omitted.

Let $Cat_R^{\mathbb{Z}_n}$ denote the category of small R-linear $\mathbb{Z}_n$-graded categories. For $C \in \mathrm{A_\infty Cat_R^{\mathbb{Z}_n,strict}}$, set $hC \in Cat_R^{\mathbb{Z}_n}$ to be the category with the same objects and with $hom_{hC}(a, b) = H^*(hom_C(a, b))$.

Given a morphism $f : A \to B$ in $\mathrm{A_\infty Cat_R^{\mathbb{Z}_n,strict}}$, we denote by $f_* : hA \to hB$ the induced morphism in $Cat_R^{\mathbb{Z}_n}$. We say that f is a *quasi-equivalence* if f_* is an equivalence. We will say that f is a *Morita equivalence* if the induced functor $\mathcal{D}(B) \to \mathcal{D}(A)$ is an equivalence where $\mathcal{D}(A)$ denotes the derived category of A. We denote by $Hmo^{\mathbb{Z}_2}$ the homotopy category of $A_\infty Cat_R^{strict,\mathbb{Z}_2}$ with respect to Morita equivalences, this is a pointed category with the zero object: the category 0 with one object and zero endomorphism complex.

The category $\mathrm{A_\infty Cat_R^{\mathbb{Z}_n,strict}}$ is cocomplete, see [11]. Furthermore, it has a model category structure [10]. This by left Bousfield localization yields a Morita model structure with weak equivalences Morita equivalences.

We will specifically need cofiber squares:

$$\begin{tikzcd} A \arrow[r, "I"] \arrow[d] & B \arrow[d, "P"] \\ 0 \arrow[r] & C, \end{tikzcd}$$

where I is a cofibration. Explicitly, a cofibration is a strict A_∞ functor, which is injective on objects and on hom sets is a cofibration of the corresponding complexes (in the projective model structure on complexes). Over a field, this means that on hom sets the functor is a degree-wise split injection. A cofiber square as above represents the homotopy cofiber.

Definition 2.1. By a ***short exact sequence*** in $A_\infty Cat_R^{strict,\mathbb{Z}_2}$ we mean a sequence $A \xrightarrow{I} B \xrightarrow{P} C$ which fits into a cofiber square as above.

As explained by Keller, an explicit example for such short exact sequences comes from Drindfeld dg-quotients.

2.1. **Simplicial sets.** We will need basic theory of simplicial sets, particularly the notion of simplex category, for the definition of bundles of A_∞ categories. We denote by Δ the simplex category:

- The set of objects of Δ is $\mathbb{N}$.
- $\hom_\Delta(n, m)$ is the set of non-decreasing maps $[n] \to [m]$, where $[n] = \{0, 1, \ldots, n\}$, with its natural order.

A simplicial set X is a functor

$$X : \Delta^{op} \to Set.$$

$X(n)$ is the set of n-simplices of X. Δ^d will denote a particular simplicial set: the standard representable d-simplex, with

$$\Delta^d(n) = hom_\Delta(n, d).$$

Morphisms of simplicial sets are given by natural trasnformations of functors and this forms a category denoted by $sSet$.

Definition 2.2. For X a simplicial set, $\Delta(X)$ will denote the ***simplex category of*** X. This is the category such that:

- The set of objects $\operatorname{obj}\Delta(X)$ is the set of simplicial maps:

$$\Sigma : \Delta^d \to X, \quad d \geq 0.$$

- Morphisms $f : \Sigma_1 \to \Sigma_2$ are commutative diagrams in $sSet$:

$$\begin{tikzcd} \Delta^d \arrow[r] \arrow[dr, "\Sigma_1"] & \Delta^n \arrow[d, "\Sigma_2"] \\ & X. \end{tikzcd} \tag{2.3}$$

Given a morphism $f : X \to Y$ of simplicial sets, we set $\Delta f : \Delta(X) \to \Delta(Y)$ to be the functor defined on objects by $\Delta f(\Sigma) = f \circ \Sigma$. If $m : \Sigma_1 \to \Sigma_2$ is a morphism in $\Delta(X)$:

$$\begin{tikzcd} \Delta^k \arrow[r, "m"] \arrow[dr, "\Sigma_1"] & \Delta^d \arrow[d, "\Sigma_2"] \\ & X, \end{tikzcd}$$

then obviously the diagram below also commutes:

$$\begin{tikzcd} \Delta^k \arrow[r, "m"] \arrow[dr, "h_1"] & \Delta^d \arrow[d, "h_2"] \\ & Y, \end{tikzcd}$$

where $h_i = \Delta f(\Sigma_i) = f \circ \Sigma_i$, $i = 1, 2$. And so the latter diagram determines a morphism $\Delta f(m) : h_1 \to h_2$ in $\Delta(Y)$. Clearly, this determines a functor Δf as needed.

Definition 2.4. Let X be a Kan complex, and C any category. Let $F_i : \Delta(X) \to C$, $i = 0, 1$ be functors. A ***concordance*** of F_0 to F_1 is a functor $\widetilde{F} : \Delta(X \times I) \to C$ such that $\widetilde{F}|_{\Delta(X\times\{0\})} = F_0$ and $\widetilde{F}|_{\Delta(X\times\{1\})} = F_1$.

3. Definition of categorified algebraic K-theory groups of R

We need to impose some algebraic finiteness conditions on our A_∞ categories, otherwise any K-theory we construct would be trivial due to the Eilenberg swindle, the same reason that K-theory of infinite dimensional vector bundles is trivial. A natural requirement is that our A_∞ categories be smooth and proper, or saturated in the sense of Kontsevich-Soibelman [9], as this is a dualizability condition, see [21]. In the $\mathbb{Z}_n$-graded setting, the analogue of this condition is more complex, we will instead work with a less restrictive notion, which is not too hard to check for geometric examples.

Set $\mathcal{A} = \mathcal{A}_R \subset A_\infty Cat_R^{\mathbb{Z}_2, strict}$ to be the full subcategory whose objects are finite type, where the latter is defined as follows. Set $w\mathcal{A}_R \subset \mathcal{A}_R$ to be subcategory with the same objects and morphisms Morita equivalences.

Definition 3.1. We say that $B \in A_\infty Cat_R^{\mathbb{Z}_2, strict}$ is ***finite type*** if the following conditions are satisfied.

- B is Morita equivalent to an A_∞ algebra, whose underlying chain complex is projective and finitely generated over R in each degree.
- The Hochschild homology complex of B over R is quasi-isomorphic to a complex (still 2-periodic) that is projective and finitely generated over R in each degree.

Definition 3.2. Let X be a simplicial set. A ***bundle of*** A_∞ ***categories over*** X is a functor $F : \Delta(X) \to w\mathcal{A}_R$. F will be called a ***bundle functor over*** X.

Suppose we have bundle functors over X, fitting into a sequence of natural transformations of functors to $\mathcal{A}_R$ (as opposed to $w\mathcal{A}_R$):

$$F_0 \to F_1 \to F_2, \tag{3.3}$$

such that for each object $\Sigma \in \Delta(S^k)$ the resulting sequence in $\mathcal{A}_R$ is a short exact sequence. Then we call $F_0 \to F_1 \to F_2$ a ***short exact sequence***.

Definition 3.4. We define $K^{Cat}(X, R)$ to be the quotient of the free abelian group generated by the set of concordance classes $[F]$ of functors $F : \Delta(X) \to w\mathcal{A}_R$, by the subgroup generated by

$$\{[F_0] - [F_1] + [F_2] \,|\, \text{there is a short exact sequence } F_0 \to F_1 \to F_2\}.$$

This is clearly contravariantly functorial in X so if $f : X \to Y$ is a simplicial map then we get a group homomorphism $f^* : K^{Cat}(Y, R) \to K^{Cat}(X, R)$ defined on generators by $f^*([F]) = [F \circ \Delta f]$.

Remark 3.5. As classical algebraic K-theory groups, this is covariantly functorial in R by "extension of scalars".

Furthermore, it is a homotopy cofunctor in X, as by definition of concordance it is clear that if f is homotopic to g then $f^* = g^*$. In particular if X_0 is homotopy equivalent to X_1 then $K^{Cat}(X_0, R) = K^{Cat}(X_1, R)$.

Set $S^k_{simp} = \Delta^k / \partial \Delta^k$ for $\Delta^k / \partial \Delta^k$ the natural quotient Kan complex.

Definition 3.6. For $k > 0$, we define

$$K^{Cat}_k(R) = \ker\{(\Delta^0 \to S^k_{simp})^* : K^{Cat}(S^k_{simp}, R) \to K^{Cat}(*, R)\},$$

and define

$$K^{Cat}_0(R) = K^{Cat}(\Delta^0, R).$$

Remark 3.7. The above definition of the groups $K^{Cat}_k(R)$ is meant to be very direct, and amenable to computation. But from a theory perspective it would be great to explain why these groups are in fact the Waldhausen K-theory groups, that is homotopy groups of a spectrum associated to a certain Waldhausen category underlying $\mathcal{A}_R$. The needed category is already outlined in Toën in [20]. We take the full subcategory $\widehat{\mathcal{A}}_R \subset \mathcal{A}_R$ with objects Morita fibrant (pretriangulated and idempotent complete). To be strict this will not be a Waldhausen category but ∞-Waldhausen in the sense of Barwick [1], as for example the zero category 0 (the category with one object and zero complex as endomorphism set) is not initial in $\widehat{\mathcal{A}}_R$ but the hom-sets $hom_{\widehat{\mathcal{A}}_R}(0, A)$ are contractible when taken in the ∞-localization of $\widehat{\mathcal{A}}_R$ with respect to the Morita equivalences. The groups $K^{Cat}_k(R)$ are then exactly the groups $\pi_k(K(\widehat{\mathcal{A}}_R))$ with $K(\widehat{\mathcal{A}}_R)$ the Waldhausen K-theory spectrum. We postpone the details to a future work.

3.1. **A very basic computation.** We recall the definition of $K_0(R)$. This is the quotient of the free abelian group generated by the set of isomorphism classes of projective finitely generated R-modules, by the subgroup generated by

$$\{[A] - [B] + [C] \,|\, \text{exists an exact sequence } 0 \to A \to B \to C \to 0.\}$$

If $\mathcal{E}$ is a triangulated category, one defines $K_0(\mathcal{E})$ to be the quotient of the free abelian group, generated by the set of isomorphism classes in $\mathcal{E}$, by the subgroup generated by

$$\{[A] - [B] + [C] \,|\, \text{exists a distinguished triangle } A \to B \to C \to A[1].\}$$

It is known that $K_k(D^{perf}(Mod_R)) \simeq K_k(R)$, where Mod_R denotes the category of projective finitely generated R-modules and $D^{perf}(Mod_R)$ denotes the derived category of perfect complexes.

If $D_n(Mod_R)$ denotes the n-periodic derived category, we no longer have that $K_0(R) \simeq K_0(D_n(Mod_R))$, as there are certain corrections, see Saito [15]. However we have the following.

Theorem 3.8 (Saito [15])**.** *For n even, the natural homomorphism:*

$$\psi : K_0(D_n(Mod_R)) \to K_0(R) \tag{3.9}$$

$$\psi([V]) = \sum_{i=0}^{i=n} (-1)^i [H^i(V)], \tag{3.10}$$

is an isomorphism.

There is a group homomorphism $K^{Cat}_0(\mathbb{Z}) \xrightarrow{\phi} K_0(D_2(Mod_{\mathbb{Z}}))$ induced by $[C] \mapsto [Hoch_\bullet(C)]$ for $Hoch_\bullet(C)$ the Hochschild homology complex of C. This readily follows from the fact that the Hochschild functor $Hoch_\bullet$ is "localizing" which means in this case that ϕ takes short exact sequence $C_0 \to C_1 \to C_2$ in $\mathcal{A}_R$ is taken to an exact triangle $Hoch_\bullet(C_0) \to Hoch_\bullet(C_1) \to Hoch_\bullet(C_2) \to Hoch_\bullet(C_0)[1]$, see Keller [7, Theorem 5.2].

Corollary 3.11. *The composition map*

$$K_0^{Cat}(\mathbb{Z}) \xrightarrow{\phi} K_0(D_2(Mod_{\mathbb{Z}})) \xrightarrow{\psi} K_0(\mathbb{Z}),$$

is non-trivial.

Proof. Take C to be a finite type $\mathbb{Z}_2$-graded A_∞ algebra over $\mathbb{Z}$ such that $HH_0(C) = \mathbb{Z}$, $HH_1(C) = 0$, $HH_2(C) = \mathbb{Z}$, where $HH_\bullet(C)$ is the homology of $Hoch_\bullet(C)$. As a geometric example take $C = FH_\bullet(L_0, L_0)$ where the latter is the Floer chain algebra of the equator L_0 in S^2, then the Hochschild homology $HH_\bullet(C)$ is isomorphic to the singular homology $H_\bullet(S^2, \mathbb{Z})$, [18].

So we get

$$\psi([Hoch_\bullet(C)]) = 2[\mathbb{Z}] \neq 0 \in K_0(\mathbb{Z}) \simeq \mathbb{Z}.$$

□

A further corollary of the proof above is:

Corollary 3.12. $[\mathrm{Fuk}(S^2, \omega)]$ *represents a non-trivial element in* $K_0^{Cat}(\mathbb{Z})$*.*

One natural question is how to produce interesting elements in $K_m^{Cat}(R)$ for higher m. We will construct such elements using Hamiltonian fibrations and Floer theory.

4. Bundles of A_∞ categories from Hamiltonian fibrations

The main construction of [17] has input a smooth Hamiltonian fibration $M \hookrightarrow P \to Y$, with (M, ω) a compact monotone symplectic manifold, e.g. $(\mathbb{CP}^n, \omega_{FS})$. Denote by $Y_\bullet$ the simplicial set of smooth maps $\Delta^n \to Y$ for Δ^n also denoting the standard topological n-simplex.

The output is a functor

$$F_P : \Delta(Y_\bullet) \to wA_\infty Cat_R^{\mathbb{Z}_2, strict}, \quad (4.1)$$

[3]

that is a bundle functor over $Y_\bullet$, whose concordance class F_P is independent of auxiliary choices (these mainly consist of choices of almost complex strutures and Hamiltonian connections.)

Definition 4.2. We call a compact monotone symplectic manifold ***finite monotone type***, if its monotone Fukaya category (see [18] and also [17]) is finite type.

We then have the following corollary of [17, Theorem 6.6].

Theorem 4.3. *Let* $M \hookrightarrow P \to Y$ *be a Hamiltonian fibration with fiber* (M, ω) *finite monotone type. Then the assignment:*

$$[P] \mapsto [F_P] \in K^{Cat}(Y_\bullet, R) \quad (4.4)$$

is well defined, where $[P]$ *is the Hamiltonian isomorphism class of the fibration* P*.*

As an example, we have:

Proposition 4.5. $(\mathbb{CP}^n, \omega_{FS})$ *is finite monotone type.*

[3] In [17] we work with $R = \mathbb{Q}$ but this was only used for inverting quasi equivalences. As we don't need to do this here, the construction works over any commutative ring.

Proof. Specifically, we take a concrete model of $\mathrm{Fuk}(\mathbb{CP}^n, \omega_{FS})$ with the set of objects given by the orbit of the Clifford torus L_0 by the natural $\mathrm{Ham}(\mathbb{CP}^n, \omega)$ action. In this case, it is automatic that $\mathrm{Fuk}(\mathbb{CP}^n, \omega_{FS})$ is quasi-isomorphic to the A_∞ algebra $hom_{\mathrm{Fuk}(\mathbb{CP}^n,\omega)}(L_0, L_0)$. The Hochschild chain complex of the latter is quasi-isomorphic to the Hamiltonian Floer chain complex, see for instance [18], which in each degree is finitely and freely generated over $\mathbb{Z}$. And so $\mathrm{Fuk}(\mathbb{CP}^n, \omega_{FS})$ is finite type for any R. □

In particular, taking $Y = S^m$ and composing with the natural map

$$K^{Cat}(S^m_\bullet, R) \simeq K^{Cat}(S^m_{simp}, R) \to K^{Cat}_m(R)$$

we get:

Corollary 4.6. *For each commutative ring R, there are natural maps:*

$$\phi_m : \pi_m(B\,\mathrm{Ham}(\mathbb{CP}^n, \omega_{FS})) \to K^{Cat}_m(R). \tag{4.7}$$

And so natural maps:

$$\phi_m : \pi_m(BPU(n) \to K^{Cat}_m(R). \tag{4.8}$$

Proof. Let $\mathbb{CP}^n \hookrightarrow P \to B\,\mathrm{Ham}(\mathbb{CP}^n, \omega_{FS})$ be the tautological Hamiltonian fibration, that is the fiber bundle associated to the universal principal $\mathrm{Ham}(\mathbb{CP}^n, \omega_{FS})$ bundle over $B\,\mathrm{Ham}(\mathbb{CP}^n, \omega_{FS})$. Use the conjunction of Proposition 4.5 and Theorem 4.3 to get the needed conclusion. □

The map cl is a group homomorphism, but to show this we need to relate our definition of $K^{Cat}_k(R)$ to the Waldhausen construction, see Remark 3.7, and this will not be carried out here.

4.1. **Generalizing to G/T.** Given a compact Lie group G, the maximal coadjoint orbit G/T can be made monotone with respect to the Kirillov-Kostant-Souriau symplectic structure ω_{KKS}, which for G/T is uniquely determined up to deformation equivalence, see for instance [8]. We will from now on omit this monotone ω_{KKS} from notation, but implicitly it will always be the chosen symplectic form. If $G = PU(n)$ there is a distinguished monotone Lagrangian torus L_{GC} of $PU(n)/T$, arising as the central fiber of a Gelfand-Cetlin system, see [3, Theorem B]. This is a kind of analogue of the Clifford torus.

It is natural to expect that $Fuk(G/T)$ is finite type. In this case we similarly get homomorphisms $\pi_m(B\,\mathrm{Ham}(G/T) \to K^{Cat}_m(R)$. Note that when we talk about $Fuk(G/T)$ ideally, we are talking about a model [4] with finitely many objects up to the $\mathrm{Ham}(G/T)$ action, similarly to how we treated $Fuk(\mathbb{CP}^n)$.

For example, form the sub-category $C_{L_{GC}}$ of $Fuk(PU(n)/T)$ with objects formed by G-orbit of L_{GC}. In this case the finite type condition is forced and we get natural homomorphisms

$$\phi_{L_{GC},m} : \pi_m(BPU(n)) \to K^{Cat}_m(R),$$

by the same construction; that is getting bundles of A_∞ categories over S^m with fiber $C_{L_{GC}}$. The images of $\phi_{L_{GC},m}$ are in principle different from the images of g_m, even restricting to $BPU(n)$. This is just because we are taking different coadjoint orbits, and so are working with different fibrations over $BPU(n)$.

We can also do this for G/T (and more general coadjoint orbits) by constructing a distinguished Lagrangian $\mathcal{L}$, as the central fiber of a toric degeneration of G/T. One of the most well known setups for this is via so called string polytopes and Newton–Okounkov bodies see [4], and G/T always has a toric degeneration. For a general G it is expected that $\mathcal{L}$ is monotone, (this in principle depends on the chosen degeneration, but it is enough for us that exists one degeneration, such that $\mathcal{L}$ is monotone). For

[4] Not necessarily Morita equivalent to $Fuk(G/T)$.

example, this holds for $PU(n)/T$ as in this case $\mathcal{L}$ can be identified with L_{GC} for the toric degeneration associated to the Gelfand–Cetlin polytope.

Assuming this for the moment we get a finite type sub-category category $C_{\mathcal{L}} \subset \mathrm{Fuk}(G/T)$, with the set of objects the orbit of $\mathcal{L}$ by the $\mathrm{Ham}(G/T)$ action. This $C_{\mathcal{L}}$ is our model.

5. Hamiltonian elements in algebraic K-theory

One of the goals is to obtain "Hamiltonian elements" in classical algebraic K-theory of R. Let $M \hookrightarrow P \to Y$ be a Hamiltonian fibration, with compact monotone fiber (M, ω). Let

$$F_P : \Delta(X_\bullet) \to wA_\infty Cat_R^{\mathbb{Z}_2, strict}$$

be as in the previous section. Then the composition: $\Phi_P = Hoch_\bullet \circ F_P$ gives a functor to $wCh_R^{\mathbb{Z}_2}$, where $Ch_R^{\mathbb{Z}_2}$ is the Waldhausen (strict) category $Ch_R^{\mathbb{Z}_2}$ of 2-periodic complexes of projective finite rank R-modules. Its Waldhausen K-theory infinite loop space will be denoted by $K^{\mathbb{Z}_2}(R)$.

For concreteness specialize to $Y = B\,\mathrm{Ham}(G/T)$, and P the associated G/T bundle over $\mathrm{BHam}(G/T)$. [5] Note that by standard homotopy theory, $|\Delta(Y_\bullet)| \simeq X$, where for a category C, $|C|$ is shorthand for the geometric realization of the nerve. Then we get a map:

$$|\Phi_P| : B\,\mathrm{Ham}(G/T) \to |wCh_R^{\mathbb{Z}_2}|,$$

whose homotopy class is well defined by the fact that the concordance class of F_P is well defined.

There is a natural map $|wCh_R^{\mathbb{Z}_2}| \to K^{\mathbb{Z}_2}(R)$, by generalities of the Waldhausen construction, see [22, Remark 8.3.2]. Composing with this map we get a natural homotopy class:

$$[|\Phi_P| : B\,\mathrm{Ham}(G/T) \to K^{\mathbb{Z}_2}(R)]. \tag{5.1}$$

Proof of Theorem 1.4. Compose the natural map

$$BG \to \mathrm{BHam}(G/T),$$

and the map from (5.1) to obtain h. $\square$

6. Langlands dual algebraic K-theory elements.

It is natural to ask how Langlands duality fits into our story. Denote by $K^\vee$ the maximal compact subgroup of the Langlands dual $G_{\mathbb{C}}^\vee$ of the complexification $G_{\mathbb{C}}$. Via Theorem 1.4, we have natural maps:

$$h_{G,m} : \pi_m(BG) \to K_m^{\mathbb{Z}_2}(R),$$
$$h_{K^\vee,m} : \pi_m(BK^\vee) \to K_m^{\mathbb{Z}_2}(R).$$

If G is simply laced (e.g. $G = SU(n)$) it is not hard to see that the images of these maps are the same. In this case $G/T = K^\vee/T$ and then the images of G and $K^\vee$ in $\mathrm{Ham}(G/T)$ coincide. So we may propose a conjecture :

Conjecture 1. *For each compact Lie group G the images of $h_{G,m}$ and $h_{K^\vee,m}$ coincide.*

Furthermore, there should be "lifts", see Section 4.1:

$$\phi_{\mathcal{L},G} : \pi_m(BG) \to K_m^{Cat}(R),$$
$$\phi_{\mathcal{L},K^\vee} : \pi_m(BK^\vee) \to K_m^{Cat}(R),$$

[5] To be even more concrete, in the construction of F_P we can work with models of the type $C_{\mathcal{L}}$, Section 4.1, so that $F_P(\Sigma)$ for each Σ is a category with finitely many objects all of which are isomorphic to a distinguished object, corresponding to $\mathcal{L}$ in Section 4.1.

and we can again ask if the images coincide. For $m = 0$, this is saying that $Fuk(G/T)$ and $Fuk(K^\vee/T)$ [6] represent the same classes in $K_0^{Cat}(R)$. This assertion holds if these categories are Morita equivalent, but it is much weaker. The case $m = 0$ is then comparable to the homological mirror symmetry conjecture of Kontsevich.

The above is thematically connected to topological aspects of Langlands duality/correspondence as in [2] for example. But I don't know a direct mathematical connection. From a physical perspective of Kapustin-Witten [5], geometric/topological aspects of Langlands correspondence are expressions of the electric-magnetic duality. It would be good to understand if there are likewise some physical connotations of the maps $h_{G,m}$.

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